

DISCORD GRAND PRIX

UNIFIED MASTER ENGINE SPECIFICATION & FEATURE GUIDE

All Bot Features | Deep Mathematical Models | Physics Equations
Admin Suite | 24 Official F1 Track Profiles | 46 Commands

SYSTEM CORE SUB-SYSTEMS:

- 6-Part Garage & Upgrade System (Levels 1-20)
- Crate Unboxing, Parts Inventory & 5-Tier Rarity Offsets
- Driver Skills Training & Dynamic Track Mastery familiarization
- 1v1 Duels, High-Stakes Wagers & full 20-Car Grand Prix weekends (Free Entry)
- Qualifying Knockouts (Q1 → Q2 → Q3) & Live Telemetry
- Advanced Physics Engine (tyre wear, non-linear degradation, mechanical DNFs)
- World Driver Championship (WDC) Season Manager & automated calendar
- Modular Python Cogs architecture built with SQLite WAL-mode databases

DOCUMENT SPECIFICATION

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Database Schema: Active Build V2 | Target: discord.py 2.x API

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1. ALL GAMEPLAY & RPG FEATURES

DISCORD GRAND PRIX is an advanced, high-performance Formula 1 esports and team management engine built for Discord. Combining deep RPG progression, realistic telemetry physics equations, a full World Driver Championship (WDC) calendar management system, an item economy with crate unboxing, and automated server activity rewards — it delivers the ultimate racing simulation experience.

Feature Subsystem	Capabilities & Functionality Overview
RPG Driver Training	Train 6 Driver Skills (Pace, Qual, Wet, Consistency, Aggression, Overtaking) with /train. Each skill ranges from 1 - 100. Training costs 400¢ per session.
6-Part Garage & Damage Engine	Upgrade & manage Engine, Aerodynamics, Tyres, ERS, Reliability, & Pit Crew (Levels 1 - 20). Dynamic damage accumulates per race with realistic power loss penalties. Repair with /repair or /repairs.
Crate & Inventory System	Unbox Rookie (500¢), Pro (2,500¢), or Champion (6,000¢) crates. Drop parts across 5 rarity tiers (Common -> Legendary) with base level offsets and efficiency multipliers. Equip parts from /inventory.
Consumable Boosters Shop	Purchase tactical boosters from /shop -- Tyre Blanket Warmer (1,500¢), ERS High-Flow Injector (2,000¢), or Heavy Duty Radiator (1,200¢). Max 2 active boosters at a time.
Economy & Earning Methods	Earn credits through /daily (500¢), /work (250 - 600¢), chat activity (25¢/msg, 1,000¢/day cap), and voice channel time (15¢/min, 1,500¢/day cap).

1v1 Duels & High-Stakes Wagers	Challenge rivals to head-to-head duel races (/race) with live lap-by-lap telemetry updates, dynamic overtaking, and real-time strategy. Wager credits using /bet.
Grand Prix Events (20-Car Grid)	Admin-scheduled full GP weekends (Free entry) with Q1/Q2/Q3 qualifying, dynamic weather, Safety Car/VSC, DRS overtaking, and F1-standard points distribution.
Sprint Race Weekends	Shorter sprint format races with alternate points distribution (8-7-6-5-4-3-2-1 for Top 8). Free entry for registered drivers. Official sprint tracks marked in calendar.
Track Mastery & Practice	Run solo practice sessions (/practice) on any official F1 track. Earn pace bonus up to -0.15s/lap. Max 3 sessions per day. Costs 500¢ per session.
WDC Season Championship	Full World Driver Championship management. Create multi-round seasons, schedule GP & Sprint rounds, view live standings, and crown a World Champion.
Leaderboard System	View server-wide rankings sorted by Championship Points, Total Wins, Team Wealth, or Driver Level via /leaderboard.

2. ONBOARDING & TEAM PROFILE CREATION

/start -- Initialize Your Racing Team

- Starting Resources: 1,500¢ Credits, Driver Level 1, 0 XP, All 6 Driver Skills at 50/100, All Garage Parts at Level 1/20 (0% Damage).
- Team Name Rules: 3 - 32 characters, must be unique per server.

/profile -- View Your Team Profile Card

Renders a premium 880x540px F1-style PNG profile card featuring carbon-teal header, overall power rating, skills column, garage status, and current wealth.

3. 6-PART GARAGE & UPGRADE SYSTEM

Component	Power Weight	Race Impact
Engine	1.0x (Highest)	Raw straight-line speed. Shaves 0.05s/lap per level.
Aerodynamics	0.8x	Corner speed & downforce. Shaves 0.04s/lap per level.
Tyres	0.5x	Grip & tyre longevity. Reduces tyre wear impact.
ERS System	0.8x	Energy recovery & deployment. Shaves 0.04s/lap per level.
Reliability	0.25x	Reduces DNF / mechanical failure probability.
Pit Crew	0.25x	Reduces pit stop stationary duration.

Damage & Repair System

- Finishing Race: Engine +3-10%, Tyres +10-25% damage.
- DNF (Crash): Engine +15-30%, Tyres +30-60% damage.
- Damage Penalty: -1% car power per 10% total damage (capped at -50%).
- Repair Costs: 25¢ per 1% damage. Commands: /repairs, /repair.

4. DRIVER PERSONNEL TRAINING

Skill	Race Impact & Mechanics
Pace	Raw lap time speed. Shaves up to 2.0s/lap at max (100).
Qualifying	One-lap qualifying speed bonus. Higher = better grid positions.
Wet Skill	Performance in rain. Reduces wet weather lap time penalty significantly.
Consistency	Reduces random lap time variance and lowers mistake/lockup probability.
Aggression	Increases overtake frequency & minor pace boost, but increases crash risk and tyre wear.
Overtaking	Improves DRS pass success rate against defender's Consistency stat.

6. CRATE UNBOXING & INVENTORY SYSTEM

Crate Tier	Price	Gold Refund	Drop Chance	Rarities Available
Rookie Crate	500¢	25 - 125¢	60%	Common (70%), Uncommon (25%), Rare (5%)
Pro Crate	2,500¢	125 - 625¢	85%	Uncommon (40%), Rare (45%), Epic (12%), Legendary (3%)
Champion Crate	6,000¢	300 - 1,500¢	100%	Rare (35%), Epic (50%), Legendary (15%)

7. CONSUMABLE BOOSTERS SHOP

Booster	Price	Tactical Effect
Tyre Blanket Warmer	1,500¢	-0.15s Qualifying Lap Pace advantage
ERS High-Flow Injector	2,000¢	+1 Lap extra ERS boost during races
Heavy Duty Radiator	1,200¢	-30% engine thermal heat buildup

8. ECONOMY & EARNING CREDITS

- Daily Login Bonus: /daily awards 500¢ flat once per 24h.
- Team Job: /work awards 250 - 600¢ random once per 24h.
- Chat Messages Passive: 25¢ per message (1,000¢ daily cap).
- Voice Channel Passive: 15¢ per minute unmuted & undeafened (1,500¢ daily cap).
- GP Winner / Podium: P1: 5,000¢, P2: 3,000¢, P3: 1,500¢ (500¢ base participation).

9. RACING MODES & GRAND PRIX EVENTS

1v1 Duels & Wagers

- /race <opponent> -- Instant head-to-head racing duel with live telemetry updates.
- /bet <opponent> <amount> -- High-stakes duel where winner takes the doubled credit pot.

Grand Prix & Sprint Weekend Lifecycle (Free Entry)

1. **Registration:** Players join with /joinrace (Free entry). Leave with /leaverace (No refunds).
2. **Qualifying (Q1 → Q2 → Q3):** Knockout format qualifying determines the starting grid.
3. **Race Simulation:** Lap-by-lap simulation featuring dynamic weather radar, DRS overtaking, Safety Car/VSC deployments, and live pit strategy management (/strategy, /pit).

Points Distribution Table

Position	GP Points	Sprint Points
P1	25	8
P2	18	7
P3	15	6
P4	12	5
P5	10	4
P6	8	3
P7	6	2
P8	4	1
P9 / P10	2 / 1	--
Fastest Lap	+1 (Top 10 only)	+1 (Top 10 only)

17. ADMIN SUITE & SERVER MANAGEMENT

- **4-Tier Permission Hierarchy:** Server Owner, Discord Administrator, Admin Role members, and Game Admins.
- **Interactive Control Panels:** /admin gp and /admin sprint launch interactive panels for track selection, driver registration management, qualifying launch, and race execution.
- **Economy & Stat Controls:** Award or deduct player credits, modify driver skill levels (1-100), alter car component levels (1-20), or reset profiles.
- **Profile Deletion:** /admin deleteuser allows permanent deletion of user profiles.
- **System Maintenance:** /admin broadcast embed announcements and /admin dbbackup.

19. COMPLETE COMMAND REGISTRY (46 COMMANDS)

Category	Commands List & Quick Function
Player Commands (20 Commands)	/start -- Initialize racing team profile /profile -- View PNG team profile card & stats /garage -- View car parts & damage levels /upgradeshop -- Browse component upgrade costs /upgrade -- Upgrade car part level (1-20) /inventory -- View stored parts & equip setup /repairs -- Check damage levels & repair costs /repair -- Pay credits to repair damaged parts /train -- Train driver skill (+1 level, 400¢) /practice -- Solo track practice for Track Mastery /daily -- Claim 500¢ daily login bonus /work -- Perform team odd-job for 250-600¢ /crate -- View loot crate odds & rewards /open -- Unbox Rookie, Pro, or Champion crate /shop -- View tactical consumable boosters /booster -- Purchase single-use race booster /race -- Challenge user to a 1v1 racing duel /bet -- Challenge user to a credit wager duel /leaderboard -- View server standings & rankings /help -- Display interactive game guide
Race Event Commands (7 Commands)	/joinrace -- Register for upcoming Grand Prix (Free) /leaverace -- Leave upcoming GP /grid -- View driver registration list & grid /standings -- View current WDC points standings /results -- View results of last completed GP /strategy -- Configure starting strategy & tyres /pit -- Schedule pit stop for next lap

Admin Suite Commands (13 Commands)	<code>/admin gp</code> -- Open Grand Prix Admin control panel <code>/admin sprint</code> -- Open Sprint Race Admin control panel <code>/admin setstat</code> -- Set driver skill or garage part level <code>/admin give</code> -- Grant credits to a player profile <code>/admin remove</code> -- Deduct credits from a player profile <code>/admin resetprofile</code> -- Reset profile to defaults <code>/admin deleteuser</code> -- Permanently delete user profile & DB rows <code>/admin resetstandings</code> -- Reset server WDC points to 0 <code>/admin broadcast</code> -- Post announcement embed to channel <code>/admin dbbackup</code> -- Backup SQLite database file <code>/admin addadmin</code> -- Grant Game Admin permission <code>/admin removeadmin</code> -- Revoke Game Admin permission <code>/admin listadmins</code> -- List active Game Admins
Season Commands (6 Commands)	<code>/season create</code> -- Create new WDC Season <code>/season add_race</code> -- Add GP round to season calendar <code>/season add_sprint_race</code> -- Add Sprint round to calendar <code>/season calendar</code> -- View & manage season calendar UI <code>/season end</code> -- Conclude season & crown Champion <code>/season cancel</code> -- Cancel active WDC season

20. DATABASE SCHEMA & ARCHITECTURE

Built on SQLite with WAL-mode concurrency and foreign keys. Modular architecture split into 5 Cogs:

- **AdminCog (cogs/admin.py)**: Admin controls, season management, GP panels.
- **GarageCog (cogs/garage.py)**: Onboarding, profile cards, garage upgrades, inventory, repairs.
- **EconomyCog (cogs/economy.py)**: Daily/work economy, crates, shop boosters, driver training, leaderboards, bets.
- **RacingCog (cogs/racing.py)**: 1v1 duels, GP lifecycle, Q1/Q2/Q3 qualifying, telemetry views.
- **SimulatorCog (cogs/simulator.py)**: Track mastery practice sessions, help system.